FCN Final Complete Notes

Expanded revision notes based on lecturer tips

Chapters 6, 7, 8, 9 and 10

This is the improved version

The previous PDF was too compressed. This version expands the five chapters into proper final-revision notes with topic probability, definitions, comparisons, calculation steps, worked examples, and practice questions.

Exam Probability Ranking

Probability	Topics	Reason
Very High	IPv4 calculation, subnetting, network/host/broadcast, usable range, VLSM/FLSM	Lecturer tips mention IPv4 calculation and VLSM/FLSM. Calculation topics are common final-paper questions.
Very High	TCP vs UDP, TCP three-way handshake, TCP termination, port numbers	Lecturer tips mention TCP session establishment/termination and TCP/UDP.
High	IPv6 representation and compression; IPv4 to IPv6 migration	Lecturer tips mention IPv6, compression of IP address, and migration techniques.
High	DNS and DHCP	Directly listed by lecturer and also important Chapter 10 services.
Medium-High	Application layer protocols and email protocols	Likely theory/comparison/port number questions.
Medium	Public/private IP, unicast/broadcast/multicast, IP and MAC delivery	Often short-answer or objective questions.

Study order: 1) IPv4 subnetting and VLSM, 2) TCP/UDP and handshake, 3) IPv6 compression and migration, 4) DNS/DHCP/email protocols, 5) public/private IP and message delivery.

Contents

1. IPv4 Addressing

IPv4 is a 32-bit logical address used to identify a device on an IP network. It is written in dotted-decimal format, for example 192.168.10.25. An IPv4 address contains a network portion and a host portion. The subnet mask or prefix length tells which bits belong to the network and which bits belong to the host.

Term	Meaning	Example
IPv4 address	Unique logical address assigned to a host/interface.	192.168.10.25
Subnet mask	Identifies network portion and host portion.	255.255.255.0
Prefix length	Number of 1 bits in the subnet mask.	/24
Network address	First address of the subnet; identifies the network itself.	192.168.10.0/24
Broadcast address	Last address of the subnet; sends to all hosts in that subnet.	192.168.10.255/24
Usable host range	Addresses that can be assigned to devices.	192.168.10.1 - 192.168.10.254
Default gateway	Router interface IP used to leave the local network.	192.168.10.1

Exam sentence: IPv4 is a 32-bit hierarchical address made up of a network portion and a host portion. A subnet mask or prefix length is used to identify the network and host portions.

Common Prefix and Subnet Mask Table

Prefix	Subnet mask	Block size	Usable hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

Memory formula: Block size = 256 - interesting octet. Usable hosts = 2^host bits - 2. Minus 2 because network address and broadcast address cannot be assigned to normal hosts.

DNS server = Domain Name System server
etc: when type www.google.com
DNS will change it to a IP address

2. Public, Private and Reserved IPv4 Addresses

Public IPv4 addresses are globally unique and routable on the internet. Private IPv4 addresses are used inside local networks and are not directly routable on the public internet. NAT is normally used when private hosts need internet access.

Network Address Translation } - used by a router to translate private IP address into a public IP address so devices inside a local network can access the internet.

Type	Range	Use
Private Class A	10.0.0.0 - 10.255.255.255 / 10.0.0.0/8	Large private networks
Private Class B	172.16.0.0 - 172.31.255.255 / 172.16.0.0/12	Medium private networks
Private Class C	192.168.0.0 - 192.168.255.255 / 192.168.0.0/16	Small/home/private networks
Loopback	127.0.0.0/8	Testing local host, for example 127.0.0.1
APIPA/link-local	169.254.0.0/16	Automatically assigned IP Address when DHCP fails
Multicast (Message delivery method where one senders sends data to a selected groups of receivers)	224.0.0.0 - 239.255.255.255	One-to-many delivery group

DHCP: Dynamic Host Configuration Protocol = A network protocol that automatically gives IP configuration to a device when it joins a network

Likely question: Identify whether an IP address is public or private. 192.168.1.5 is private. 8.8.8.8 is public. 172.20.1.1 is private. 172.32.1.1 is public because the private Class B range stops at 172.31.

3. IPv4 Message Delivery Options

Delivery option	Meaning	Example
Unicast	One sender to one receiver.	PC sends a packet to one printer.
Broadcast	One sender to all hosts in the local network. (hosts: Any device connected to a network.)	ARP request or DHCP Discover in IPv4. (ARP: Address Resolution Protocol: a protocol used to find MAC address when its IP is known)
Multicast	One sender to selected group members.	Streaming or routing-protocol group traffic.

MAC = Media Access Control, used to identify a device inside a local network

Important difference: broadcast reaches every host in the broadcast domain, but multicast only reaches hosts that joined the multicast group. IPv6 does not use broadcast; it uses multicast instead.

4. IPv4 Calculation and Subnetting

IPv4 calculation is the most likely final-exam calculation topic. You must be able to find subnet mask, network address, first usable address, last usable address, broadcast address, number of subnets and usable hosts.

Steps to Calculate Subnet Information

1. Identify the prefix length, for example /26.
2. Convert the prefix to subnet mask, for example /26 = 255.255.255.192.
3. Find the interesting octet. It is the octet that is not 255 and not 0. For /26, the interesting octet is 192.
4. Calculate block size: 256 - 192 = 64.
5. List subnet ranges by adding block size: 0, 64, 128, 192.
6. Find which range contains the given IP address.
7. Network address = first address in range. Broadcast = last address before next range. Usable range = network + 1 to broadcast - 1.

Worked Example 1: 192.168.10.77/26

Step	Working	Answer
1	/26 means subnet mask 255.255.255.192.	Mask = 255.255.255.192
2	Block size = 256 - 192 = 64.	Block = 64
3	Subnets: 192.168.10.0, .64, .128, .192.	77 is inside .64 subnet
4	Range is 192.168.10.64 - 192.168.10.127.	Network = 192.168.10.64
5	First usable = network + 1.	192.168.10.65
6	Last usable = broadcast - 1.	192.168.10.126
7	Broadcast = last address of range.	192.168.10.127
8	Host bits = 32 - 26 = 6. Usable hosts = 2^6 - 2.	62 usable hosts

Final answer format: Network = 192.168.10.64/26; first usable = 192.168.10.65; last usable = 192.168.10.126; broadcast = 192.168.10.127; usable hosts = 62.

Borrowed Bits and Host Bits

Borrowed bits are bits taken from the host portion to create more subnets. Host bits are the remaining bits used for host addresses inside each subnet.

Formula	Use
Number of subnets = 2^borrowed bits	Use when the question asks how many subnets are created.
Number of addresses per subnet = 2^host bits	Includes network and broadcast addresses.
Number of usable hosts = 2^host bits - 2	Excludes network and broadcast addresses.
Host bits = 32 - prefix length	Example: /27 has 5 host bits.
Borrowed bits = new prefix - original prefix	Example: class C /24 changed to /27 means borrowed bits = 3.

Worked Example 2: A /24 network is subnetted to /27

Item	Working	Answer
Borrowed bits	27 - 24	3 bits
Number of subnets	2^3	8 subnets
Host bits	32 - 27	5 host bits
Usable hosts per subnet	2^5 - 2	30 usable hosts
Subnet mask	/27	255.255.255.224
Block size	256 - 224	32

5. FLSM vs VLSM

FLSM means Fixed Length Subnet Mask. Every subnet uses the same subnet mask and has the same number of host addresses. VLSM means Variable Length Subnet Mask. Different subnets can use different subnet masks based on different host requirements.

Comparison	FLSM / Classful style	VLSM
Subnet size	All subnets same size.	Subnets can be different sizes.
Efficiency	May waste many IP addresses.	More efficient because each subnet gets what it needs.
Difficulty	Easier to calculate.	More difficult; must sort largest to smallest.
Use case	Simple networks with equal host needs.	Real networks with different department sizes.
Exam clue	Question says equal subnet size.	Question gives different host requirements.

FLSM = 把一个 subnet address (192.168.1.0/26) 的 /26，一样大小给别 department.

Department	IP数量
Admin	10
IT	50
Finance	20
HR	15

都会被分配 /26 的 siz

usable
192.68.1.1 — 192.68.1.62
192.68.1.65 — 192.68.1.126
192.68.1.129 — 192.68.1.190
192.68.1.193 — 192.68.1.254

FLSM Example

Question: 192.168.1.0/24 needs 4 equal subnets. Find the new prefix and ranges.

1. Need 4 subnets, so borrow 2 bits because 2^2 = 4.
2. Original prefix /24 + 2 borrowed bits = /26.
3. Subnet mask = 255.255.255.192.
4. Block size = 64.
5. Subnets: 192.168.1.0/26, 192.168.1.64/26, 192.168.1.128/26, 192.168.1.192/26.

Subnet	Network	First usable	Last usable	Broadcast
1	192.168.1.0	192.168.1.1	192.168.1.62	192.168.1.63
2	192.168.1.64	192.168.1.65	192.168.1.126	192.168.1.127
3	192.168.1.128	192.168.1.129	192.168.1.190	192.168.1.191
4	192.168.1.192	192.168.1.193	192.168.1.254	192.168.1.255

VLSM Example

Question: Use 192.168.10.0/24 for departments requiring 60 hosts, 30 hosts, 12 hosts and 2 hosts.

> VLSM rule: Sort requirements from largest to smallest. Choose the smallest subnet that can fit each requirement.

Requirement	Need addresses	Prefix	Usable hosts	Allocated subnet
60 hosts	Need at least 62 addresses including network/broadcast	/26	62	192.168.10.0/26
30 hosts	Need at least 32 addresses	/27	30	192.168.10.64/27
12 hosts	Need at least 16 addresses	/28	14	192.168.10.96/28
2 hosts	Need at least 4 addresses	/30	2	192.168.10.112/30

Subnet	Network	Usable range	Broadcast
60 hosts	192.168.10.0/26	192.168.10.1 - 192.168.10.62	192.168.10.63
30 hosts	192.168.10.64/27	192.168.10.65 - 192.168.10.94	192.168.10.95
12 hosts	192.168.10.96/28	192.168.10.97 - 192.168.10.110	192.168.10.111
2 hosts	192.168.10.112/30	192.168.10.113 - 192.168.10.114	192.168.10.115

VLSM marking tip: Always show largest-to-smallest ordering, prefix selection, network address, usable range and broadcast address. This makes your working clear.

6. IPv6 Addressing and Compression

IPv6 uses 128-bit addresses written in hexadecimal. It was introduced because IPv4 has limited address space and the internet has too many devices. IPv6 also supports address autoconfiguration and reduces dependence on NAT.

Item	IPv4	IPv6
Address size	32 bits	128 bits
Writing format	Decimal dotted notation	Hexadecimal hextets separated by colon (0→9) (A-F)
Example	192.168.1.10	2001:0db8:0000:1111:0000:0000:0000:0200
Broadcast	Supported	Not used; uses multicast
NAT	Commonly used	Less needed because huge address space
Address assignment	Static or DHCP	Static, SLAAC, DHCPv6

手动设置,IP固定

Stateless Address Autoconfiguration

IPv6里面的一种自动配置IP address的方法

SLAAC

- Device creates its own IP address
- No Dynamic Host Configuration Protocol server needed
- Uses router, router gives network prefix
- Less control for admin
- simple automatic IPv6 setup

DHCPv6

- DHCPv6 gives the IPv6 address
- Needs DHCPv6 server
- Works with router information
- More control for admin
- Managed network, company, school

IPv6 Address Compression Rules

1. Omit leading zeros in each hextet. Example: 0db8 becomes db8, 0000 becomes 0, 0200 becomes 200.
2. Use double colon :: to replace one continuous sequence of all-zero hextets.
3. Double colon can be used only once in one IPv6 address. Otherwise, the address becomes ambiguous.

Worked Example: Compress IPv6 Address

Step	Address
Original	2001:0db8:0000:1111:0000:0000:0000:0200
Remove leading zeros	2001:db8:0:1111:0:0:0:200
Use :: for longest zero sequence	2001:db8:0:1111::200

GUA ip address

> Exam warning: Do not use :: more than once. 2001:db8::1111::200 is invalid because the receiver cannot know how many zero hextets each :: represents.

IPv6 Address Types

IPv6 type	Typical prefix / example	Meaning
Global Unicast Address (GUA)	2000::/3	Publicly routable IPv6 address, similar to public IPv4. GUA = public IPv6 address
Link-Local Address (LLA)	FE80::/10	Used only on local link. Required for IPv6-enabled interfaces. Routers do not forward it. LLA = private IPv6 address only, cannot go to Internet
Multicast	FF00::/8	One-to-many delivery to group members. One sender to a selected group
Loopback	::1	Used by a host to send traffic to itself. Sends back the data to ownself
Unspecified	::	Means no address; used before an address is assigned. No specific IPv6 assigned

> IPv6 GUA structure usually has three parts: global routing prefix, subnet ID and interface ID. In many LAN designs, /64 is used, meaning the first 64 bits identify the network/subnet and the last 64 bits identify the interface.

7. IPv4 to IPv6 Migration Techniques

IPv4 and IPv6 coexist because the internet cannot change from IPv4 to IPv6 instantly. There are three main migration techniques: dual stack, tunneling and translation.

Technique	Meaning	Simple example	Key word
Dual stack	Device runs IPv4 and IPv6 at the same time.	A PC has both 192.168.1.10 and 2001:db8::10.	Both protocols
Tunneling IPv6 packet → 包进IPv4 packet里面 → 经过IPv4 network → 拿回出来变IPv6	IPv6 packet is encapsulated inside IPv4 packet to cross IPv4-only network.	IPv6 islands communicate across IPv4 internet.	IPv6 inside IPv4
Translation / NAT64 NAT64 translate IPv6 to IPv4 IPv6 to IPv4	Translates IPv6 packet to IPv4 packet or vice versa.	IPv6-only client accesses IPv4-only server.	Convert protocol

Exam answer sentence: Dual stack allows IPv4 and IPv6 to run simultaneously on the same device; tunneling carries IPv6 traffic inside IPv4 packets; NAT64 translation allows IPv6-only and IPv4-only devices to communicate.

8. Network Layer and IP/MAC Delivery

The network layer is OSI Layer 3. It provides addressing, encapsulation, routing and de-encapsulation so that end devices can exchange data across networks. IPv4 and IPv6 are the main network layer protocols.

Process	Meaning
Addressing	IP addresses identify source and destination end devices. → Devices that are used by users to send or receive data in a network
Encapsulation	Transport layer segment is wrapped with an IP header to become a packet.
Routing	Routers choose the best path to reach the destination network.
De-encapsulation	Destination host removes the IP header and passes data upward.

OSI = 7 layer

发送数据时: Application → Presentation → session → Transport → Internet → Data link → Physical link
接收数据时: Physical layer → Data Link → Internet → Transport → Session → Presentation → Application

Application: 应用程序产生数据
Presentation: 处理数据格式,加密
Session: 建立和管理通信会话
Transport: 负责可靠传送数据(TCP/UDP)
Internet: 负责IP地址和路由

Data Link: 处理Mac地址和数据帧

Physical Layer: 把数据变成信号(0→1)发送

IP Address vs MAC Address

Item	IP address	MAC address
Layer	Layer 3 Network layer	Layer 2 Data Link layer
Purpose	Identifies network and host across networks.	Identifies device/interface on local network segment.
Changes?	Can change when device moves network.	Usually fixed on NIC.
Example	192.168.1.10	00-1A-2B-3C-4D-5E
Used by	Routers for routing between networks.	Switches for forwarding frames in LAN.

Same Network vs Different Network Delivery

Situation	Destination IP	Destination MAC in frame
Same network	Final destination host IP	MAC address of destination host
Different network	Final destination host IP	MAC address of default gateway/router

Important: The destination IP address usually stays the same from source to final destination, but the destination MAC address changes at every hop because each local link needs a new Layer 2 frame.

9. Transport Layer: TCP and UDP

The transport layer manages communication between applications on different hosts. It tracks conversations, segments and reassembles data, adds transport headers, and uses port numbers to identify applications.

Feature	TCP (Transmition Control Protocol)	UDP (User Datagram Protocol)
Connection	Connection-oriented; establishes a session first.	Connectionless; no session establishment.
Reliability	Reliable; uses acknowledgements and retransmission.	Best-effort; no guarantee.
Ordering	Segments are sequenced and reassembled in order.	Data is processed in the order received.
Flow control	Supported.	Not supported in the same way.
Speed/overhead	More overhead, slower but reliable.	Less overhead, faster but unreliable.
Common uses	Web, email, file transfer where accuracy matters.	DNS query, DHCP, streaming, VoIP, online games.

- TCP make sure the data is delivered correctly. If data is wrong, TCP will send again.

- UDP sends data quickly, but it does not guarantee that the data will arrive

Lecturer tip says “UTP and TCP”, but based on Chapter 9 it most likely means UDP and TCP. UTP is a cable type, while TCP/UDP are transport layer protocols.

Port Numbers

Port Number is the identifier for specific application or service running on a device

Port range	Name	Meaning
0 - 1023	Well-known ports	Reserved for common services such as HTTP, DNS, SMTP.
1024 - 49151	Registered ports	Assigned to specific applications/vendors.
49152 - 65535	Dynamic/private/ephemeral ports	Usually temporarily assigned to client applications.

Common Well-Known Ports

Protocol	Port	Transport	Use
FTP Data	20	TCP	File transfer data
FTP Control	21	TCP	File transfer control
SSH	22	TCP	Secure remote login
Telnet	23	TCP	Unsecure remote login
SMTP	25	TCP	Send email
DNS	53	TCP/UDP	Name resolution
DHCP Server	67	UDP	Server listens for DHCP
DHCP Client	68	UDP	Client receives DHCP
TFTP	69	UDP	Simple file transfer
HTTP	80	TCP	Web
POP3	110	TCP	Retrieve email
IMAP	143	TCP	Retrieve/sync email
HTTPS	443	TCP	Secure web

10. TCP Session Establishment and Termination

TCP Three-Way Handshake

Before TCP sends application data, it establishes a session using a three-way handshake. The purpose is to synchronize sequence numbers and confirm both devices are ready to communicate.
同步序列数据

Step	Direction	Flag	Meaning
1	Client -> Server	SYN	Client requests to start a TCP session and sends initial sequence number. [Clients ask to start a connection]
2	Server -> Client	SYN + ACK	Server accepts and acknowledges client sequence number; server sends its own sequence number. [Server accepts and replies]
3	Client -> Server	ACK	Client acknowledges server sequence number. Connection established. [Clients confirm the connections]

Memory: SYN, SYN-ACK, ACK. Exam sentence: TCP uses the three-way handshake to establish a reliable connection before data transfer begins.

TCP Session Termination => End the connection after data transfer is finished

TCP normally terminates a connection gracefully using FIN and ACK messages. Either side can start closing the session.

Step	Direction	Flag	Meaning
1	Host A -> Host B	FIN	Host A wants to close its side of the connection.
2	Host B -> Host A	ACK	Host B acknowledges the FIN.
3	Host B -> Host A	FIN	Host B also closes its side.
4	Host A -> Host B	ACK	Host A acknowledges. Connection closed.

TCP reliability summary: numbering/sequence numbers, acknowledgement, retransmission, same-order delivery and flow control help TCP provide reliable delivery.

Application Layer是 user application 和 network 之间的接口 (Application 不会自己处理底层网络, 它们会通过 Application Layer 来使用网络服务

11. Application Layer Protocols

The application layer provides the interface between user applications and the underlying network. In the TCP/IP model, OSI application, presentation and session layer functions are grouped under the application layer.

Protocol	Full name	Port	Purpose
HTTP	Hypertext Transfer Protocol	TCP 80	Transfers web pages and web content.
HTTPS	HTTP Secure	TCP 443	Secure web communication using encryption.
FTP	File Transfer Protocol	TCP 20/21	Transfers files with control and data connection.
TFTP	Trivial File Transfer Protocol	UDP 69	Simple file transfer, no authentication, low overhead.
DNS	Domain Name System	TCP/UDP 53	Converts domain names to IP addresses.
DHCP	Dynamic Host Configuration Protocol	UDP 67/68	Automatically assigns IP settings.
SMTP	Simple Mail Transfer Protocol	TCP 25	Sends email between clients/servers and mail servers.
POP3	Post Office Protocol v3	TCP 110	Downloads email from server to client.
IMAP	Internet Message Access Protocol	TCP 143	Accesses and synchronizes email on server.

Presentation layer functions: data formatting, compression and encryption/decryption. Session layer functions: creates, maintains and restarts dialogs between applications.

12. Email Protocols

Email normally uses different protocols for sending and receiving. SMTP is mainly used for sending email. POP3 and IMAP are mainly used for receiving/accessing email.

Protocol	Used for	Simple explanation	Port
SMTP	Sending email	Transfers mail from client to mail server and between mail servers.	TCP 25
POP3	Receiving email	Downloads email to local device; usually simpler.	TCP 110
IMAP	让自己的email在不同设备同步 Receiving/<u>syncing email</u> etc:电话 delete email,电脑的email也会被delete	Keeps email on server and synchronizes across devices.	TCP 143

Exam difference: POP3 is suitable when you mainly use one device and want to download email. IMAP is better when you use many devices because messages remain on the server and stay synchronized.

Domain Name System

13. DNS Protocol

DNS translates human-readable domain names into IP addresses. Users remember names such as www.example.com, but computers need IP addresses to communicate.

Record type	Meaning	Example use
A	Maps a name to an IPv4 address.	example.com -> 93.184.216.34
AAAA	Maps a name to an IPv6 address.	example.com -> 2001:db8::1
NS	Identifies authoritative name server.	Which server manages the domain.
MX	Mail exchange record.	Which mail server receives email for domain.

Basic DNS Process

1. User enters a domain name in browser.
2. Client checks local DNS cache.
3. If not found, client asks configured DNS resolver.
4. DNS resolver searches through DNS hierarchy if needed.
5. DNS returns the matching IP address.
6. Client uses the IP address to contact the destination server.

Important: DNS usually uses UDP port 53 for normal queries because it is fast. TCP port 53 can be used for larger responses or zone transfers.

14. DHCP

DHCP automatically assigns IPv4 settings to hosts. It reduces manual configuration errors and is commonly used for end-user devices. Static addressing is usually used for routers, switches, servers and printers.

DHCP provides	Meaning
IPv4 address	Host address assigned from DHCP pool.
Subnet mask	Allows host to know its local network.
Default gateway	Router address used to reach other networks.
DNS server address	DNS resolver used for name resolution.
Lease time	How long the address is reserved for the client.

DORA Process

Step	Message	Direction	Meaning
1	Discover	Client -> Broadcast	Client searches for DHCP server.
2	Offer	Server -> Client	Server offers an IP configuration.
3	Request	Client -> Server	Client requests offered address.
4	Acknowledgement	Server -> Client	Server confirms lease and settings.

Ports: DHCP server uses UDP port 67 and DHCP client uses UDP port 68.

15. Practice Questions and Quick Answers

1. What is the difference between public and private IP address?

Answer: Public IP is globally routable on the internet. Private IP is used inside local networks and usually needs NAT to access the internet.

2. State three IPv4-to-IPv6 migration techniques.

Answer: Dual stack, tunneling and translation/NAT64.

3. Compress 2001:0db8:0000:0000:0000:ff00:0042:8329.

Answer: 2001:db8::ff00:42:8329

4. What is the difference between TCP and UDP?

Answer: TCP is reliable and connection-oriented. UDP is connectionless, faster and best-effort.

5. What is the TCP three-way handshake?

Answer: SYN, SYN-ACK, ACK.

6. What does DNS do?

Answer: DNS resolves domain names into IP addresses.

7. What does DHCP do?

Answer: DHCP automatically assigns IP address, subnet mask, default gateway and DNS information to hosts.

8. Given 192.168.5.100/27, find block size.

Answer: /27 = 255.255.255.224. Block size = 256 - 224 = 32.

9. Given 192.168.5.100/27, find network and broadcast.

Answer: Subnets are .0, .32, .64, .96, .128. 100 is in .96 range, so network = 192.168.5.96 and broadcast = 192.168.5.127.

10. Difference between FLSM and VLSM?

Answer: FLSM uses the same mask for all subnets. VLSM uses different masks based on host requirements.

Last-Minute Formula Sheet

Need to find	Formula / method
Host bits	32 - prefix length
Usable hosts	2^host bits - 2
Total addresses	2^host bits
Borrowed bits	New prefix - original prefix
Number of subnets	2^borrowed bits
Block size	256 - interesting octet
Network address	First address in subnet range
First usable	Network address + 1
Broadcast	Last address in subnet range
Last usable	Broadcast - 1

Common Subnet Masks

/24	255.255.255.0	254 usable hosts
/25	255.255.255.128	126 usable hosts
/26	255.255.255.192	62 usable hosts
/27	255.255.255.224	30 usable hosts
/28	255.255.255.240	14 usable hosts
/29	255.255.255.248	6 usable hosts
/30	255.255.255.252	2 usable hosts

Final study priority: spend most time on IPv4 subnetting/VLSM and TCP/UDP because those are calculation + comparison topics. Then memorise IPv6 compression, migration techniques, DNS, DHCP and email protocols.